SOFTWARE ENGINEERING LAB (UE24CS341A)

SOFTWARE ENGINEERING

PROJECT : SMARTLABEQUIPMENTSLOTRESERVATIONPORTAL

High-concurrencyequipment reservation, calibrationsafetyguards, and accessgovernance system

TEAM 1

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PROJECT OVERVIEW & PROBLEM STATEMENT

Customer / Target Users

- Students & Researchers: Transparent, conflict-free advance equipment booking.
- Lab Technicians: Manage calibration, maintenance states, check-in/out.
- Lab Administrators: User role governance, policy and audit log review.

Our Proposed Solution

A high-concurrency web application offering real-time equipment catalog exploration, automated conflict detection via PostgreSQL range exclusion constraints, and role-based operational workflows.

Problem Statement

University research labs face severe scheduling conflicts, double-booking race conditions, and uncalibrated equipment usage when relying on manual sign-up sheets or basic spreadsheets.

Unmonitored late returns also prevent peers from accessing critical apparatus for scheduled laboratory sessions.

Project Key Objectives

- Zero Overlap: Enforce PostgreSQL GiST exclusion constraints & row locks.
- Calibration Guard: Block uncalibrated equipment reservations with HTTP 422.
- Return & Late Tracking: Track late return incidents and penalty counters.
- End-to-End Governance: JWT HttpOnly cookie storage, rate limit, audit trails.

PROPOSED SOLUTION & SYSTEM ARCHITECTURE

MODULE	FUNCTIONALITY & CONSTRAINTS
Authentication & RBAC	Dual JWT in HttpOnly cookies, bcrypt, STUDENT/TECH/ADMIN roles.
Catalog & Availability	Live search, lab filters, slot grid, and smart alternative suggestions.
Reservation Engine	Row-level lock, max 2-hr duration, 7-day window, calibration guard.
Return & Late Tracking	Technician verification, automated is_late flag, penalty counters.
Admin Governance	User role management, rate limiter (100 req/min), audit trails.

Technology Stack:

- Frontend: React 18, Vite, Tailwind CSS, Lucide Icons, Axios
- Backend: Node.js, Express REST API (Controllers -> Services -> Repositories)
- Database: PostgreSQL 15 with btree_gist extension (3NF Relational Model)
- Testing & CI: Jest, Supertest, Docker Compose, GitHub Actions Automated CI

USER INTERFACE LAYER (React 18 SPA)

Vite Dev Server (Port 5173) ¢ Axios with HttpOnly Credentials

↓ REST API

EXPRESS APPLICATION LAYER (Port 5000)

Rate Limiter (100 req/min) ¢ AuthMiddleware ¢ RoleGuard (/api/v1)

↓ Layered Architecture

BUSINESS & RESERVATION ENGINE

Pessimistic Row Lock (SELECT FOR UPDATE) ¢ Calibration Guard

↓ pg Pool Client

POSTGRESQL 15 RELATIONAL DATABASE

3NF Schema ¢ EXCLUDE USING gist(tstzrange) ¢ Append-Only Audit

TEAM CONTRIBUTIONS, SE PRACTICES & CONCLUSION

Agile Delivery & Evaluation

KSHAMA - Team Lead & Repo Owner — 26 Pts

Sprint 1: Seed data setup, catalog API, calibration guard (HTTP 422).

Sprint 2: Admin user management UI/API, audit log viewer, refresh tokens.

MANJUNATHA H J - Database & Engine — 26 Pts

Sprint 1: PostgreSQL 3NF schema, btree_gist exclusion, row-level locks.

Sprint 2: Technician equipment status, tech dashboard, 100-user load test.

MOHAMMED ARFAN ASGAR - Auth & DevOps — 26 Pts

Sprint 1: Monorepo skeleton, GitHub Actions CI, JWT auth & RBAC middleware.

Sprint 2: Student reservation cancellation, technician return, rate limiting.

N. LAKSHANYAA - Frontend & Testing — 26 Pts

Sprint 1: React app shell, Login & modal UI, My Reservations view, durability test.

Sprint 2: Sprint 2 unit test suite, integration test suite, demo video, docs.

CONCLUSION & FINAL OUTCOME:

A high-concurrency, enterprise-grade labreservation platform delivered following modern Agile SE practices.

All 28 planned issues completed with 100% CI pass rate & zero double-booking verified under 100 concurrent requests.

28 Issues | 104 Story Pts